

1.4 Matrices

Further Pure 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A matrix represents a linear transformation. Matrix multiplication composes transformations in reverse order of application; determinants reveal area scale and invertibility. Eigenvectors retain direction under transformation, with eigenvalue scale factor. An inverse exists only when determinant is nonzero.

Worked example: follow the reasoning

For $A = \text{diag}(2, 3)$, $A(1, 0) = (2, 0)$ and $(1, 0)$ is an eigenvector with eigenvalue 2; determinant 6 scales area by six.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 For $A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$, find eigenvalues and determinant, and describe transformation of x and y coordinates. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Eigenvalues 2 and -1 [1]; $\det A = -2$ [1]; x doubles while y is reflected across x -axis [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.